- **Chapter 6**
- **Counting**

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Combinatorics

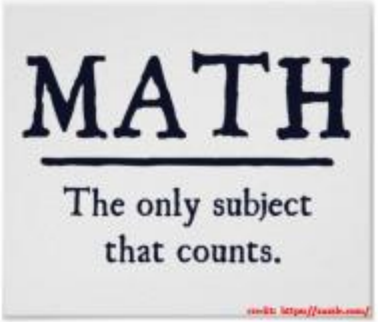
Combinatorics is the study of arrangement of objects

The Art of Counting (enumerative combinatorics)

Why Count?

Counting and Combinatorics:

- Calculate the chances of a component failure
- Analyze how long a program takes to finish
- Very useful in designing algorithms



Why Count?

- How many bit strings of length n are there?
- How many paths between two nodes on the Internet?
- How many valid IP addresses are there?
- How many steps are required to do sorting?
- How many valid 6 character passwords are possible?
- How many ways are there to buy 13 different bagels from a shop that sells 17 types?
- How many bit strings of length 11 contain a streak of one type of bit of exact length 7?
- How many ways can a hiring service match 13 jobs to 17 applicants?
- How many arrangements are there of a deck of cards?

Counting Techniques

We will learn basic counting techniques to answer this type of questions and apply them in applications mentioned above

- Sum Rule and Product Rule
- The Complement principle
- Inclusion-exclusion principle
- Pigeon-hole principle
- Permutations and Combinations

Sum Rule

Suppose you have 9 long sleeves shirts and 8 short sleeves shirts

How many choices do you have for a shirt?

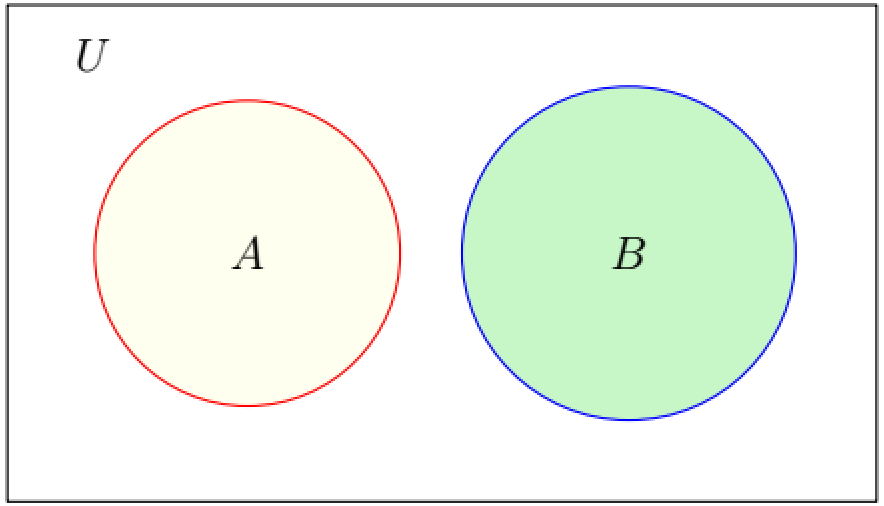
$$9 + 8 = 17$$

Sum Rule

Sum Rule

If a task can be done either in one of n_1 ways or in one of n_2 different ways, then there are $n_1 + n_2$ ways to do the task

Sum Rule



Sum Rule

If A and B are disjoint sets, then

$$|A \cup B| = |A| + |B|$$

Sum Rule

I have 32 students in one section and 35 students in the other section. Suppose I give the grade A to one student.

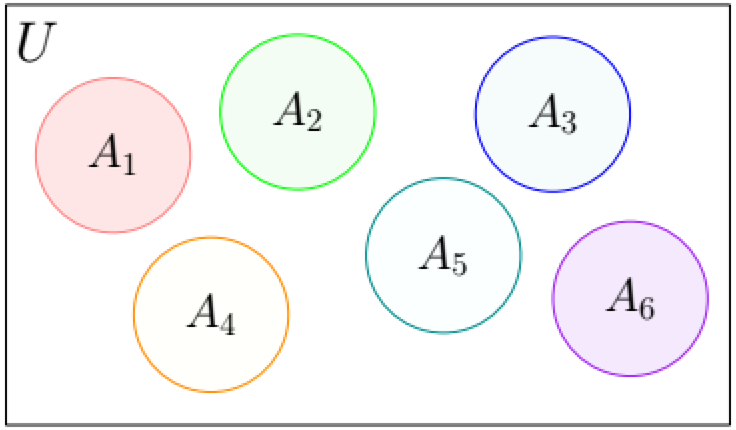
How many choices do I have in total?

Suppose you have to choose a project from 4 software development projects or from 5 research projects.

How many choices do you have?

The sum rule is easy, but identifying the sets or constructing them requires some thinking and trial and error

Generalized Sum Rule



Sum Rule

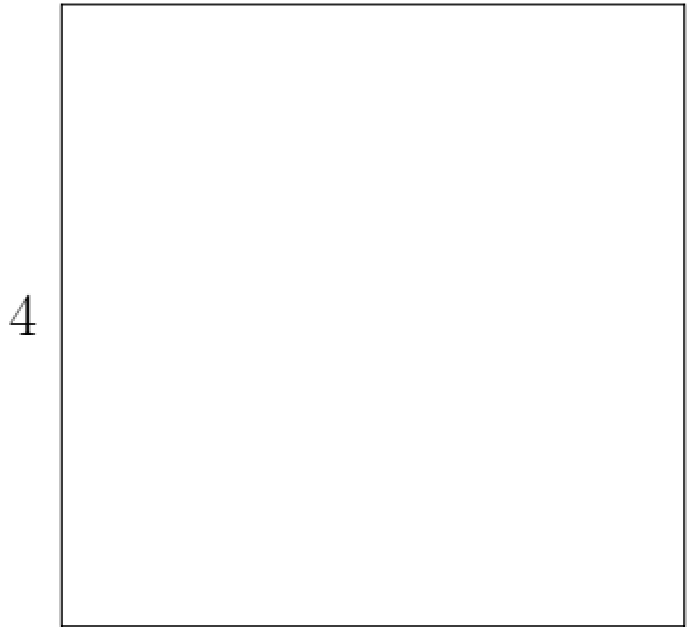
If $A_1, A_2, \ldots, A_n$ are **disjoint sets**, then

$$\left|\bigcup_{i=1}^{n} A_i\right| = \sum_{i=1}^{n} |A_i|$$

Sum Rule

Divide $4'' \times 4''$ square into 16 unit squares by parallel lines

ICP 11-1 How many squares are there in total?



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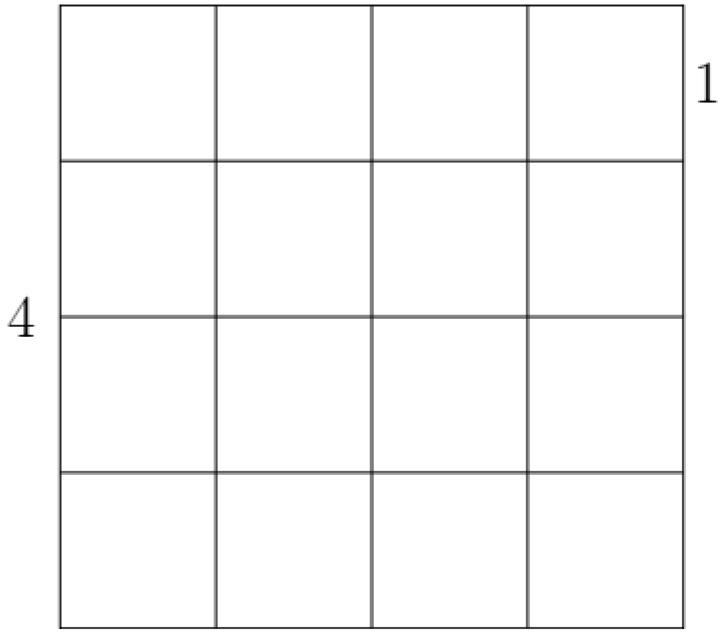


A_k : Number of squares of length k

Total number of squares are

$|A_1| + |A_2| + |A_3| + |A_4|$

$|A_1| = 16 \quad |A_2| = 9 \quad |A_3| = ? \; |A_4| = ?$

Product Rule

Suppose you have 9 shirts and 6 pairs of pants

How many choices do you have for an outfit?

$$9 \times 6 = 54$$

Product Rule

Suppose you have 9 shirts, 6 pairs of pants and 4 ties

How many choices do you have for an outfit?

$$9 \times 6 \times 4 = 216$$

Product Rule

Product Rule

Suppose a procedure can be broken down into two successive tasks.

- If there are n_1 ways to do the first task and
- for each way of doing the first task, there are n_2 ways to do the second task,

then there are $n_1 \times n_2$ ways to do the procedure

Product Rule

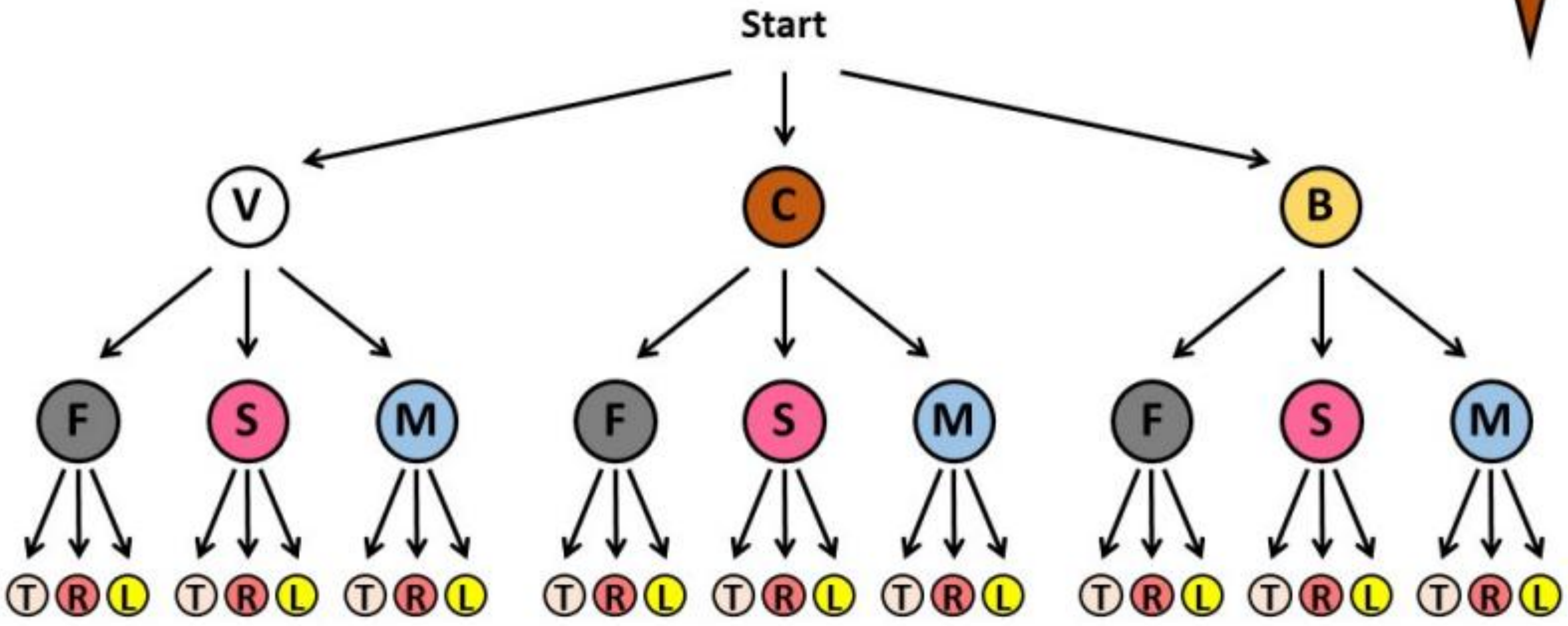
Flavours
Vanilla
Chocolate
Banana

Toppings
Flake
Sprinkles
Marshmallows

Sauce
Toffee
Raspberry
Lemon

We can **map** the possibilities.

What would happen to the total amount of possibilities if we can choose a sauce as well?



To find the total possibilities we can **multiply** each independent option.

$3 \times 3 \times 3 = 27$ total possibilities

source: https://www.goteachmaths.co.uk/

Product Rule

Set theoretic version of the Product Rule

$$|A \times B| = |A| \times |B|$$

In general,

$$|A_1 \times A_2 \times \ldots \times A_n| = |A_1| \times |A_2| \times \cdots \times |A_n|$$

Product Rule

How many ways to arrange the letters A and B ?

Break the procedure of arranging A and B into two successive tasks

1. choose first letter $\triangleright$ 2 ways to do it
2. choose second letter $\triangleright$ For each way for task-1, 1 way to do it

Number of ways to arrange A and B is (2×1)

$AB \qquad BA$

Product Rule

ICP 11-3 How many ways to arrange the letters A, B, and C ?

Break the procedure of arranging A, B, and C into 3 successive tasks

1. choose first letter ▷ 3 ways to do it
2. choose second letter ▷ For each way for task-1, 2 ways to do it
3. choose third letter ▷ For each way for task-1 & 2, 1 way to do it

$$ABC, \quad ACB, \quad BAC, \quad BCA, \quad CAB, \quad CBA$$

Product Rule

A license plate contains 3 letters and 4 digits.

How many different license plates are possible?

L	E	B	5	7	0	0

1st Place – 26 choices
2nd Place – 26 choices
3rd Place – 26 choices
4th Place – 10 choices
5th Place – 10 choices
6th Place – 10 choices
7th Place – 10 choices

$$(26)^3 \times (10)^4$$

Product Rule

ICP 11-7 How many 5 digits Postal Codes are there?

0	8	8	5	4

1st Place – 10 choices ▷ (0,1,2,3,4,5,6,7,8,9)
2nd Place – 10 choices ▷ (0,1,2,3,4,5,6,7,8,9)
3rd Place – 10 choices ▷ (0,1,2,3,4,5,6,7,8,9)
4th Place – 10 choices ▷ (0,1,2,3,4,5,6,7,8,9)
5th Place – 10 choices ▷ (0,1,2,3,4,5,6,7,8,9)

$$(10)^5$$

Product Rule

ICP 11-8 How many 5 digits Postal Codes are there with no repetition?

0	8	9	3	7

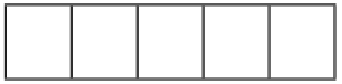
1st Place –	10 choices	$\triangleright$ (0,1,2,3,4,5,6,7,8,9)
2nd Place –	9 choices	$\triangleright$ (1,2,3,4,5,6,7,8,9)
3rd Place –	8 choices	$\triangleright$ (1,2,3,4,5,6,7,9)
4th Place –	7 choices	$\triangleright$ (1,2,3,4,5,6,7)
5th Place –	6 choices	$\triangleright$ (1,2,4,5,6,7)

$$(10 \times 9 \times 8 \times 7 \times 6)$$

Product Rule

ICP 11-9 How many passwords can be made with the following rules?

- Can contain digits and case sensitive English letters
- Must be 5 to 7 characters long
- Must begin with a letter



P_5 := set of length 5 passwords

P_6 and P_7 are similarly defined

$$\begin{aligned} |P_5| &= 52 \times 62^4 \\ |P_6| &= 52 \times 62^5 \\ |P_7| &= 52 \times 62^6 \end{aligned}$$

P_5, P_6, P_7 make partition of all valid passwords

A 'valid' password in exactly one of P_5, P_6, or P_7

By the sum rule, total number of valid passwords $= |P_5| + |P_6| + |P_7|$

Complement Principle

To count the elements in a set A, find $|\overline{A}|$

Suppose I roll a red and blue dice!

In how many outcomes the dice show different values?

A_i = set of outcomes with

red die shows i and the blue die shows something else

S = set of outcomes where the dice show different values

$$|S| = \left|\bigcup_{i=1}^{6} A_i\right| = \sum_{i=1}^{6} |A_i| = \sum_{i=1}^{6} 5 = 30$$

Complement Principle

To count the elements in a set A, find $|\overline{A}|$

Suppose I roll a red and blue dice!

In how many outcomes the dice show different values?

S = set of outcomes where the dice show different values

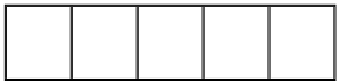
B = set of outcomes where the dice agree

$|S| = 36 - |B|$

$$|S| = 36 - 6 = 30$$

Complement Principle

$\boxed{\textbf{ICP 11-10}}$ How many 5 digits postal codes are there with at least one repetition?



How many 5 digits postal codes are there?

$$(10)^5$$

How many 5 digits postal codes are there with no repetition?

$$10 \cdot 9 \cdot 8 \cdot 7 \cdot 6$$

$$|\text{ All codes }| - |\text{ codes with no repetition }|$$

$$\underbrace{10^5}_{\text{All}} - \underbrace{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6}_{\text{codes with no repetition}}$$

Inclusion-Exclusion Principle

I have 32 students in one section and 35 students in the other section. Suppose I give the grade A to one student.

How many choices do I have in total?

$$\text{Number of choices} \;=\; 32 + 35$$

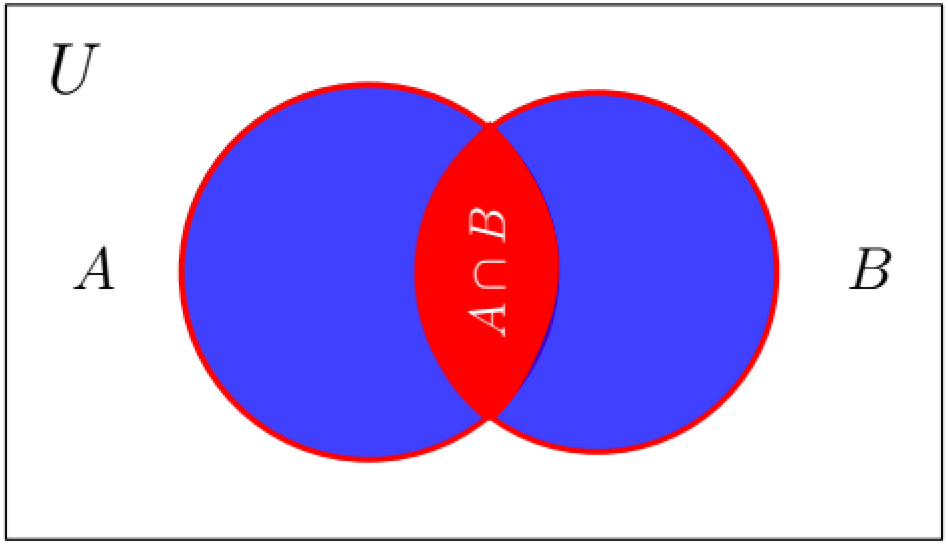
I have 32 students in Disc. Math. and 35 students in Prog. course. Suppose I give the grade A to one student.

How many choices do I have in total?

$$\text{Number of choices} \;=\; 32 + 35$$

Students in both courses are counted twice

Inclusion-Exclusion Principle



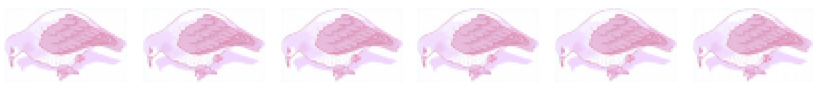
Inclusion-Exclusion Principle

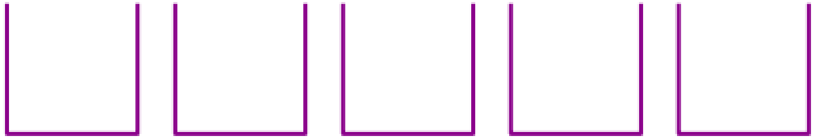
$$|A \cup B| = |A| + |B| - |A \cap B|$$

$\boxed{\textbf{ICP 11-11}}$ Let $A = \{2, 4, 7, 9\}$ and $B = \{1, 3, 7, 8, 9, 5\}$.

What is $|A \cup B|$?

Pigeonhole Principle

If there are more pigeons 

than there are pigeon holes, 

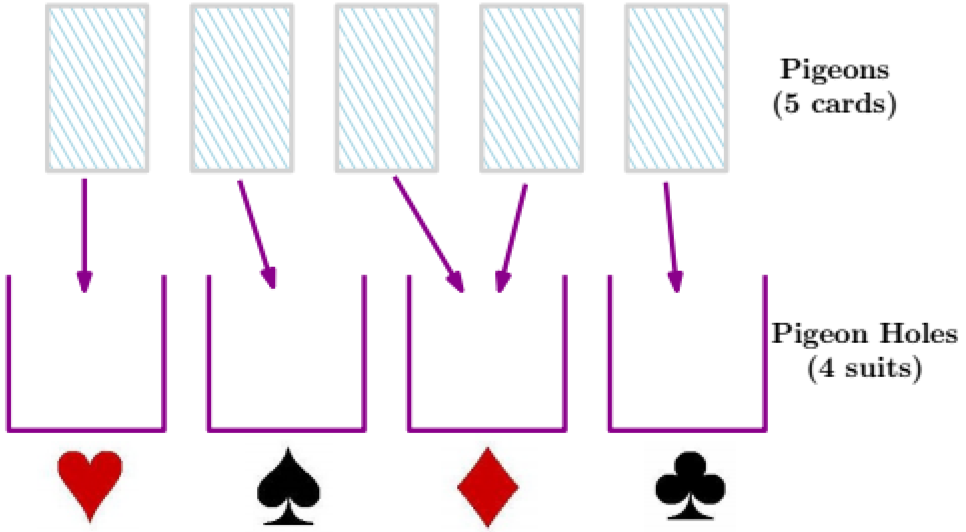
then some pigeon hole must have more than one pigeons

This is obvious! Less obvious! Where would this ever be useful?

- Probability of collision by hash functions
- Proof that there cannot be a lossless compression of some size

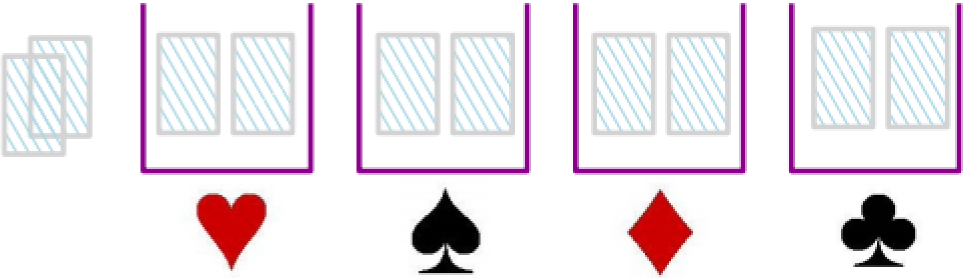
Pigeonhole Principle

Any set of 5 cards must have at least 2 cards of the same suit



Pigeonhole Principle

Any set of 10 cards must have at least how many of the same suit



$$\text{Number of cards of the same suit} \quad \geq \left\lceil \frac{10}{4} \right\rceil$$

Generalized Pigeonhole Principle

$\boxed{\textbf{ICP 11-16}}$ Prove that in any set of 13 people at least two have the same birth month

- Pigeons: The 13 peoples' birth months
- Pigeon holes: The 12 months in a year

Generalized Pigeonhole Principle

6 exam time slots per day in the exam week (7 days)

08:00AM 10:00AM	10:00AM 12:00PM	12:00PM 02:00PM	02:00PM 04:00AM	04:00PM 06:00AM	06:00PM 08:00PM

- No two simultaneous exams in a room
- 300 courses exams have to be done in one week

How many rooms are needed at the minimum?

$$\left\lceil \frac{300}{42} \right\rceil = 8$$

General Pigeonhole Principle

If $\mathbf{n}$ pigeons are placed into $\mathbf{h}$ holes,

then some hole must have at least $\left\lceil \frac{\mathbf{n}}{\mathbf{h}} \right\rceil$ pigeons

Generalized Pigeonhole Principle

ICP 11-17

Five departments, each with 100 students.

How many students should be selected so that from some department at least 6 students are selected?

$$\left\lceil \frac{n}{5} \right\rceil = 6 \quad \text{or} \quad \frac{n}{5} > 5 \quad \text{or} \quad n > 25$$

Generalized Pigeonhole Principle

1000 people go from LHR to ISB in 22 buses each with 60 seats

$\boxed{\textbf{ICP 11-18}}$ Some bus must carry more than at least x people

- Pigeons: The 1000 filled seats
- Pigeon holes: The 22 buses

$\boxed{\textbf{ICP 11-19}}$ Some bus must have at least y empty seats

- Pigeons: The empty seats
- Pigeon holes: The 22 buses

Permutation

How many ways to arrange the letters A, B, C, and D ?

$$4 \times 3 \times 2 \times 1$$

How many 5 digits codes are there with no repetition ?

$$10 \times 9 \times 8 \times 7 \times 6$$

How many ways to arrange n letters ?

$$n \times (n-1) \times (n-2) \times \cdots\cdots\cdots \times 3 \times 2 \times 1$$

How many r digits codes are there with no repetition from n characters ?

$$n \times (n-1) \times (n-2) \times \cdots \times (n-(r-1))$$

Permutation

A permutation or arrangement of n objects is an ordering of the objects

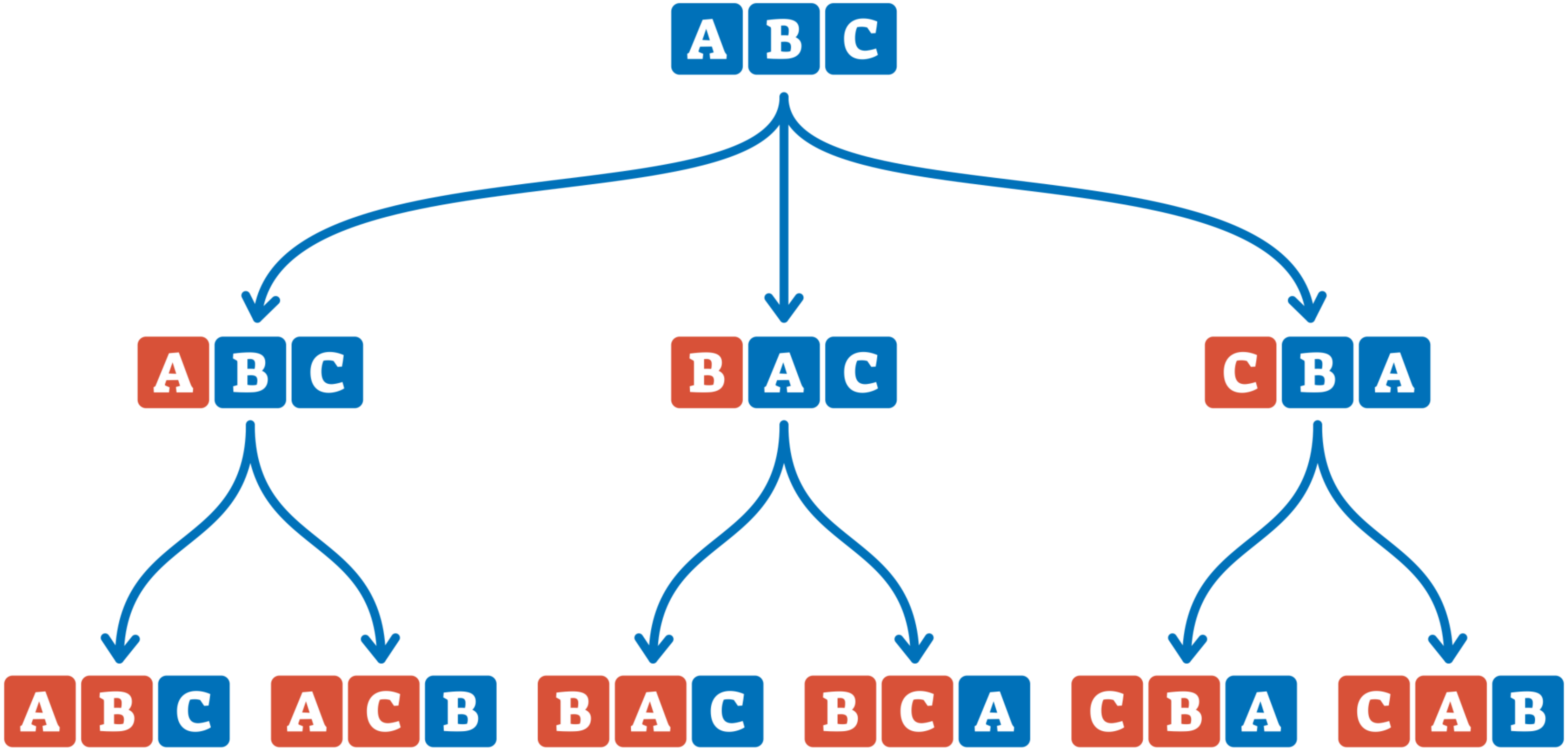
The number of permutations of n objects is $n!$

An r-permutation is an ordering or arrangement of r out of n objects

The number of r-permutations of n objects is

$$n(n-1)(n-2)\cdots(n-(r-1))$$

PERMUTATION



Permutation

$P(n, r)$: Number of r-permutation of n objects

Number of ways to permute, order or arrange r out of n objects

$$\overbrace{\boxed{}\boxed{}\boxed{}\boxed{\cdots}\boxed{}\boxed{}\boxed{}}^{r \text{ boxes}}$$

$$P(n, r) = n(n-1)(n-2)\cdots(n-(r-1))$$

$$= \frac{n!}{(n-r)!}$$

Ordered Versus Unordered

I have 5 students, $\{A, B, C, D, E\}$ and I give the prizes \$500 and \$300 to two students.

How many choices do I have?

Order of selection matters here, A, B is different than B, A

$$P(5, 2)$$

Ordered Versus Unordered

I have 5 students, $\{A, B, C, D, E\}$ and I give the grade A to two students.

How many choices do I have?

Order of selection does not matter, A, B is the same as B, A

$\{A, B\}$	AB, BA	Number of 2 letter arrangements from $\{A, B, C, D, E\}$
$\{A, C\}$	AC, CA	
$\{A, D\}$	AD, DA	
$\{A, E\}$	AE, EA	$P(5,2) = \frac{5!}{3!} = 20$
$\{B, C\}$	BC, CB	
$\{B, D\}$	BD, DB	But these are ordered pairs
$\{B, E\}$	BE, EB	Each unordered pair is listed twice
$\{C, D\}$	CD, DC	
$\{C, E\}$	CE, EC	Each unordered pair is listed 2! times
$\{D, E\}$	DE, ED	Number of unordered pairs $= 20/2! = 10$

Ordered Versus Unordered

$\boxed{\textbf{ICP 11-24}}$ How many subsets of size 5 are there for the set $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$?

Number of ordered 5 length sequences that can be formed from $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

$$P(10, 5) = \frac{10!}{(10-5)!} = 30240$$

Number of arrangements/ordering of 5 digits is

$$P(5, 5) = 5! = 120$$

Each subset is counted 120 times. Number of (unordered) subsets

$$\frac{P(10, 5)}{5!} = \frac{10!}{(10-5)!5!} = \frac{30240}{120} = 252$$

Combinations

An r-combination is a (unordered) subset of size r of n objects

The number of r-combinations of n objects is

$$\frac{P(n,r)}{r!} = \frac{n!}{(n-r)!r!} = \binom{n}{r}$$

$\binom{n}{r}$: n choose r (binomial coefficient)

(the number of r-subsets of an n-element set)

Combinations

Number of r-combinations of n objects is $\binom{n}{r} = \dfrac{n!}{(n-r)!r!}$

Number of 0-subsets of $\{a,b,c,d\} = \binom{4}{0} = \dfrac{4!}{(4-0)!0!} = 1$

Number of 1-subsets of $\{a,b,c,d\} = \binom{4}{1} = \dfrac{4!}{(4-1)!1!} = 4$

Number of 2-subsets of $\{a,b,c,d\} = \binom{4}{2} = \dfrac{4!}{(4-2)!2!} = 6$

Number of 3-subsets of $\{a,b,c,d\} = \binom{4}{3} = \dfrac{4!}{(4-3)!3!} = 4$

Number of 4-subsets of $\{a,b,c,d\} = \binom{4}{4} = \dfrac{4!}{(4-4)!4!} = 1$

Combinations

ICP 11-25 How many 3-bit sequences with *two* 0's and *one* 1

- 3 possible positions to put the first 0
- 2 remaining positions to put the second 0
- 1 remaining to put the 1

$$3 \times 2 \times 1$$

$$001, 010, 100$$

Choose the 2 positions for 0's and the 1 is forced

$$\binom{3}{2} = 3$$

Combinations

ICP 11-26 How many 7-bit sequences with 3 0's and 4 1's

Choose 3 positions for 0's and the 1's are forced

$$\binom{7}{3}$$

Equivalently,

Choose 4 positions for 1's and the 0's are forced

$$\binom{7}{4}$$

Counting with Combinations

A school committee of 8 faculty members has to be formed, subject to

- The committee should have 5 and 3 members from the EE and CS dept
- EE department has 30 and CS has 20 members

How many different committees can be made?

$$\binom{30}{5} \times \binom{20}{3}$$